

Trinity

The hardware compiler where RTL and the reference model cannot drift.

Dmitrii Fedorov - Founder & CTO (publishes as Dmitrii Vasilev)
dmitrii.f@t27.ai - github.com/gHashTag/t27 - Ko Samui, Thailand
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PROBLEM

Chip teams write every design twice

RTL for silicon. A reference model for checking. Two languages, two authors - and the two drift apart constantly. Verification, not design, eats the engineering calendar.

- Teams report most of their time goes to testing and verification - Silogy (YC) puts it at roughly 70%
- Every drift bug is found late, on a workload, at the most expensive moment
- Commercial equivalence tools police the drift at five-to-six-figure per-seat prices - and still do not generate the model

SOLUTION

One spec. Four targets. Bit-exact, enforced by CI.

You write a single .t27 specification. The compiler emits synthesizable Verilog, C, Rust and an executable reference model from that one source.

- CI fails any pull request where the four disagree bit for bit on a full workload
- Drift is not policed - it is impossible by construction
- The whole flow runs on open tools: yosys, nextpnr, prjxray. No commercial EDA licence anywhere

WORKING TODAY

Proven on our own silicon path, not slideware

- A complete backpropagation training loop, generated by the compiler, runs and learns on a live Artix-7 board
- The same source passed gate-level sign-off on SKY130: 18 of 18 tests
- Ternary pipeline synthesises at 4,267 LUTs, zero DSP48 - every multiply is add/subtract over $\{-1,0,+1\}$ weights (real Yosys results; place-and-route not yet closed, so we quote no Fmax)
- Public repo: github.com/gHashTag/t27 - 100+ specs, the bit-exact gate running on every pull request

Every claim on this slide is reproducible from the public repository.

PUBLISHED

The arithmetic underneath is peer-visible

Two arXiv preprints document the number formats, including pre-registered criteria for reporting a negative result.

- arXiv:2606.05017 - GoldenFloat: a phi-derived static-split floating-point family
- arXiv:2606.09686 - an 83-format numeric catalog with bit-exact conformance vectors
- A public evidence ledger marks claims we have withdrawn ourselves - we would rather show the red jobs with the green ones

WHY NOW

Open silicon just created our first users

A generation of teams now designs chips without commercial EDA: SkyWater open PDK, Tiny Tapeout, RISC-V university programs, fabless AI startups on open flows.

- They have no \$100K/seat verification budget - and every one of them hand-writes a reference model
- Nobody serves them: commercial tools are priced for incumbents, academic tools stop at RTL
- We live in their toolchain already - same yosys, same open PDKs

MARKET

Verification is the largest cost inside a ~\$18-20B industry

Industry trackers put EDA at roughly \$18-20B a year. Verification is its largest single cost.

- Entry: hundreds of fabless AI-chip startups and open-silicon teams working in open toolchains today
- Expansion: FPGA teams inside larger companies; then commercial chip teams - the pytest path, from small projects into enterprises
- Business model to validate at SkyDeck: open core plus paid hosted CI equivalence gate per seat/design

COMPETITION

Adjacent categories, none removes the drift

- HDL generators (Chisel, SpinalHDL, Amaranth): free, mature - and emit RTL only. No model, no C, no Rust, no cross-target gate
- Commercial equivalence (Synopsys, Cadence, Siemens): proves RTL matches RTL after a change; still does not generate the model
- AI verification startups (ChipAgents, Silogy, ChipStack): help write and debug tests faster. We remove the class of bug their tests hunt
- Honest caveat: they all have customers. We do not, yet. What we alone have: spec, RTL, model and two software targets as one artifact, proven to silicon sign-off, fully open

TRACTION

Zero external users - stated plainly, with the plan

The compiler has one user: its author. That use is deep - everything on slide 4 was built through it.

- Single current focus: the first external team running a .t27 spec
- SkyDeck is the shortest path to them: the people who built Chisel and RISC-V are on campus, their students are exactly our first users
- The public repo, the CI gate and both preprints are the top of that funnel

TEAM

Solo founder - the whole stack, one pair of hands

Dmitrii Fedorov (publishes as Dmitrii Vasilev). Compiler author, number-format designer, author of both preprints.

- Took a ternary design from spec to SKY130 gate-level sign-off alone, on open tools
- Got a compiler-generated backprop loop learning on a live FPGA
- The first engineering hire is what this round funds - verification benchmarks and developer experience

THE ASK

\$1.8M pre-seed - to turn an internal compiler into a product

Target valuation \$8M. Bootstrapped to date; burn \$2,500/month; runway independent of hype.

- First engineering hire
- First 10 external teams onboarded, with public case studies
- Hosted CI equivalence gate - the first paid product
- SkyDeck Batch 23: \$210K, the Berkeley network, and the exact campus our first users study on